Literature Review

1. Explainable AI-Based Smart Irrigation (Aditya et al., 2026)

Aditya et al. (2026) published a paper in which they designed a sustainable, IoT-driven smart irrigation system strengthened with explainable artificial intelligence. Their approach combined an Isolation Forest model for anomaly detection with five different machine learning classifiers, ultimately relying on XGBoost as the core prediction engine, and used SHAP to make the model's decisions interpretable for farmers rather than treating it as a black box. A two-task architecture was designed so that the same pipeline could both classify whether irrigation was needed and estimate how long the pump should run. The dataset was collected from a real field prototype set up in Ponneri, Tamil Nadu, where soil moisture, temperature, humidity, light and rainfall were logged every fifteen minutes over ninety days, producing more than twenty-five thousand raw readings that were later condensed into around five thousand hourly records suitable for training. The results were impressive, with the XGBoost model achieving an AUC-ROC of 0.995 and an F1-score of 0.992, while inference took only a few microseconds per sample and the full sensor-to-actuator response occurred within four to six seconds. Overall water consumption dropped by roughly thirty-five percent compared to conventional scheduling. However, the authors acknowledged that outdoor sensors degrade over time and that building long-term trust in the model's decisions across multiple crops and regions remains an open challenge that future work needs to address carefully.

2. AI-Driven Irrigation Using Edge-Based Embedded Controllers (Al-Jame & Manthila, 2026)

Al-Jame and Manthila (2026) published a paper in which they proposed an edge-based smart irrigation system that keeps all AI decision-making local rather than depending on the cloud. Their design paired a Raspberry Pi 4, which ran a lightweight supervised Decision Tree model, with an STM32 microcontroller that handled real-time actuation of the pump and valves, while capacitive soil-moisture, DHT22 temperature-humidity and rain sensors supplied the input data. The dataset was gathered directly from vegetable farms in semi-arid zones over a two-month period, with more than seventeen thousand samples collected at five-minute intervals and manually labelled using agronomic expertise so the model could learn realistic irrigation timing. The results showed that the system achieved above ninety-four percent decision accuracy when compared against expert-labelled irrigation decisions, with an average inference latency of only thirty-four milliseconds and total power consumption kept under two and a half watts. Field trials also reported up to thirty-eight percent water savings and a twenty percent improvement in crop yield when measured against traditional timer-based irrigation over a six-week comparison period. Despite these strong results, the authors noted that the trained model is closely tied to the specific crops and climate conditions it was trained on, meaning it would

need retraining before being deployed elsewhere. They suggested that future versions could incorporate reinforcement learning, federated learning across multiple farms, and solar-powered operation to make the system more adaptable and self-sustaining in real agricultural settings.

3. Smart Irrigation Using Multi-Model AI and Reinforcement Learning (Soy & Salwadkar, 2024)

Soy and Salwadkar (2024) published a paper in which they combined several machine learning techniques rather than relying on a single model, aiming to capture both short-term and long-term irrigation patterns. Their system used Random Forest Regression to predict near-term soil moisture, an LSTM network to model soil-water dynamics over several days, and a Q-Learning reinforcement learning agent that learned how to reduce unnecessary irrigation events over time, all supported by an IoT wireless sensor network and edge computing for faster local decisions. The dataset consisted of real-time soil moisture, temperature and humidity readings collected from tomato and maize test plots, giving the models exposure to two different crop types with different water needs. The results were encouraging across every model: the Random Forest achieved an R^2 of 0.91 for short-term predictions, the LSTM reached an RMSE of 0.07 for five-day water dynamics, and the Q-Learning agent cut over-irrigation events by forty-five percent. The combined system reported water savings of up to thirty-eight percent alongside a twelve to eighteen percent improvement in crop yield, while edge processing reduced data transmission latency by forty-two percent and lowered communication energy use by eighteen percent. Even so, the authors pointed out that scaling this multi-model approach to much larger farms, calibrating sensors across varied soil types, and keeping the system affordable for smallholder farmers remain significant practical challenges that still need to be resolved before wide deployment.

4. Multi-Model AIoT System for Precision Water Management (Abith et al., 2026)

Abith et al. (2026) published a paper in which they proposed a multi-modal AIoT irrigation system that goes beyond simple soil-moisture thresholds by combining several data sources into one decision pipeline. Their architecture used an LSTM network to forecast soil moisture trends, integrated live weather API data for rainfall-aware planning, and applied crop-specific hydration modelling based on the FAO Penman-Monteith equation, with Linear Programming and Genetic Algorithms used afterward to optimize the final irrigation schedule. The system was built on ESP32 and Raspberry Pi hardware communicating over MQTT, and it was evaluated across a ninety-day cropping season on tomato and brinjal plots, using soil, weather and crop-growth data collected throughout the trial. The results showed approximately ninety-three percent prediction accuracy with a system response latency of just over one second, while the full case study reported nearly thirty-eight percent reduction in water consumption, a yield

improvement of ten to twelve percent, and close to a forty percent reduction in labour requirements compared to manual irrigation practices. Despite these strong outcomes, the authors were careful to note that the system had only been tested on two crop types within one climate context, and that broader validation across different crops, climates, sensor failures and data noise conditions would be necessary before the approach could be considered reliable for widespread real-world deployment across diverse farming regions.

5. Hybrid K-Means and SVM Intelligent Irrigation System (Kumar et al., 2024)

Kumar et al. (2024) published a paper in which they explored combining unsupervised clustering with supervised classification to improve irrigation decision accuracy. Their method first applied Correlation-based Feature Selection to identify the most relevant soil and humidity variables, then used K-Means clustering to group similar environmental conditions before feeding this structured information into a Support Vector Machine classifier for the final irrigation decision, comparing this hybrid approach against Random Forest and Naïve Bayes models. The dataset combined IoT-collected soil and humidity readings with historical environmental records, and notably the supporting data was made openly available through the SmartFasal platform, which adds a degree of transparency and reproducibility that many similar studies lack. The results showed that the hybrid K-Means plus SVM model achieved 98.5 percent accuracy, clearly outperforming both Random Forest and Naïve Bayes when tested on the same dataset, while the system also reported freshwater savings of up to twenty percent under the evaluated conditions. Although these results are strong, the authors acknowledged that the model had not yet been validated across a wide range of field conditions, crop types or deployment environments, and that sensor quality and data consistency would need closer attention before the system could be trusted for large-scale agricultural use. Nonetheless, the use of an open, reusable dataset makes this a particularly citable and dependable reference point.